

# Final Project

## Technology Economics

Martin Kaplánek, Matouš Fiala, Jáchym Líba

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### 1 Introduction

Large cities all over the world suffer from heavy traffic, recently many of them decided to do something about that using different precautions. Urbanists are trying to come up with ideal solution applicable to every city, which is rather complicated task. The measures to keep the traffic bearable are different; somewhere, subsidizing public transport or shared mobility might work, maybe in combination with making these transport means more reliable and preferable over cars. When this is not enough, more restrictive measures must be used. Recently, many cities implemented congestion tolls in the city centres to financially motivate drivers to switch to other transport means while not banning the entry absolutely. This "kinda" tax can be considered as an external mechanism that accounts for the externality of heavy traffic which would be otherwise not financially "visible".

IN the context of the situation described above and given that the case is highly topical around big cities, we aim to focus on one specific outcome of this policy - the intermodal switch, specifically to shared bikes. Leveraging the unique dataset of New York CitiBike provided, we outline a clear research question: After implementation of the congestion toll in New York, did the commuters switch to shared bikes?

Given the nature of the policy and nature of the data, we employ two quasi-experimental (?) research designs: Regression Discontinuity and Difference in Differences. We selected them to match the situation: the toll was implemented strictly on one day and was effective since and the toll area is strictly separated - this allows us to use both the time and spatial boundary in our methodological approach.

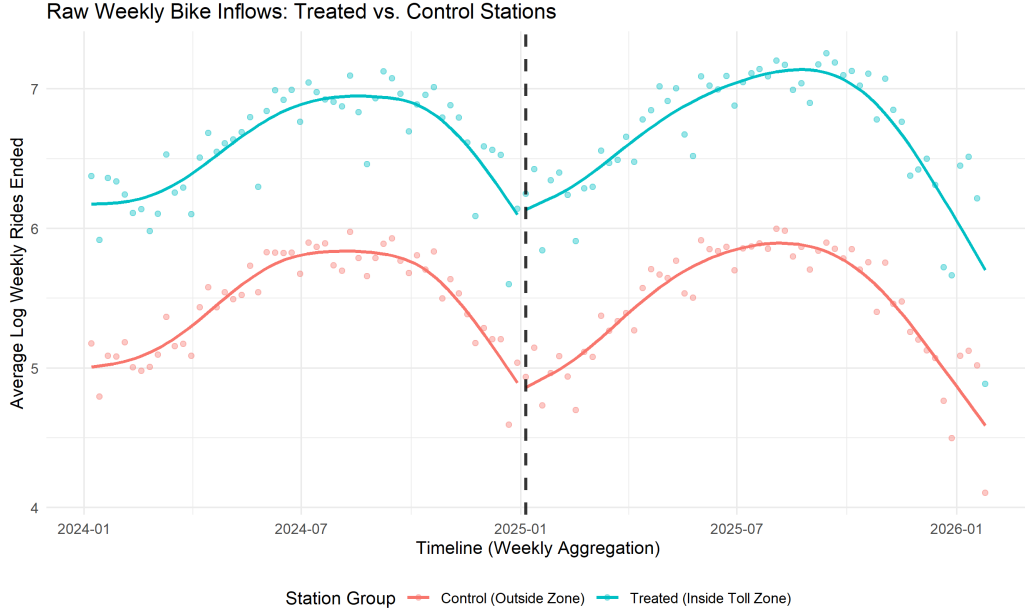


Figure 1: Enter Caption

## 2 Literature

## 3 Data

### 3.1 Difference in Differences

Before the DiD estimation, it is beneficial to view the raw data and observe possible trends. We do this using two approaches: Figure ?? presents the raw data fitted using flexible function, Figure ?? presents seasonally adjusted data. While the former suffers from extreme seasonality, which makes the potential evaluation of parallel trends hard, the latter shows only the variation the data that is left after adjusting for individual stations and both weekly and monthly characteristics (we do this adjustment by first regressing the weekly rides ended on station-month and week fixed effects; second, we obtain the residuals and use them as the adjusted values).

The raw figure shows not distinction between the group dynamics before and after the effect, we can observe that indeed the stations inside and outside the toll zone are different even before the toll, but we do not see any substantial change after its introduction. The second figure, using the adjusted residuals show some trends. First, it suggest that parallel trend assumption is satisfied, as the fitted lines are nearly parallel. Second, there is noticeable change in level of the fitted line after the toll introduction - while the treated zone increases ant once, the control groups does exactly the opposite, suggesting, there was a switch of where rides ended, consistent with intuitive expectations of the effect.

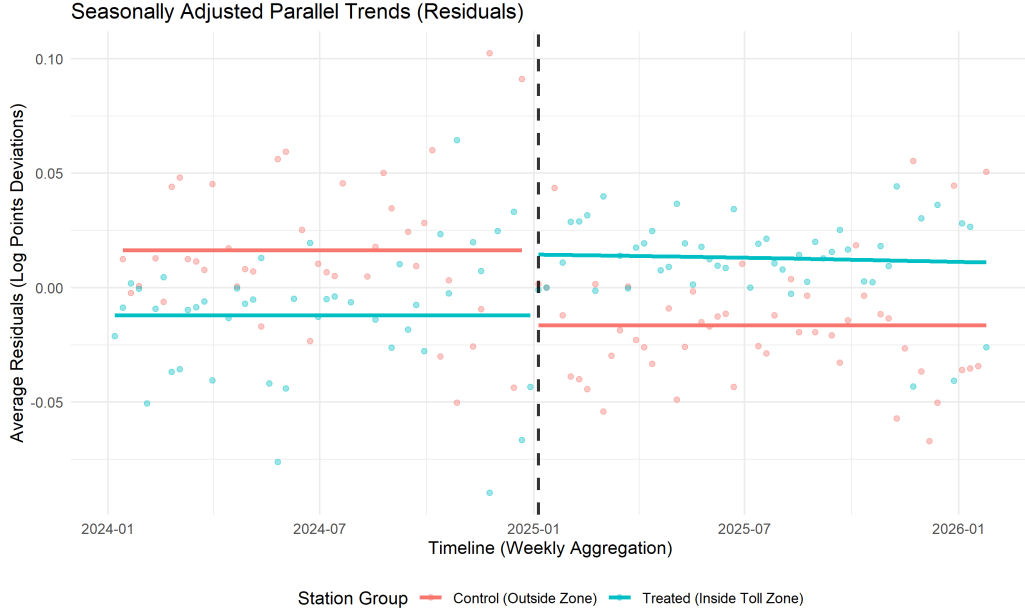


Figure 2: Enter Caption

## 4 Methodology

## 5 Results

### 5.1 Difference in Differences

We begin by presenting the vent study plots of both rides ended and net rides (...) While the net rides show not effect, there is an effect observed for rides ended. But this can be observed on both sides of the figure, suggesting a seasonality is playing a role rather that causal effect of toll introduction. Note that these event studies are estimated already using the fixed effects as in the final DiD estimation.

These visualizations suggest the underlying data might be problematic for DiD estimation, as there is effect detected even before the treatment is applied, this substantially limits the reliability of our results and analysis.

The results of preferred models, as show in Figures (...) show a statistically significant effects for some specifications - namely Rides started and ended in logs, for levels, it does not show significant effect.

## 6 Discussion

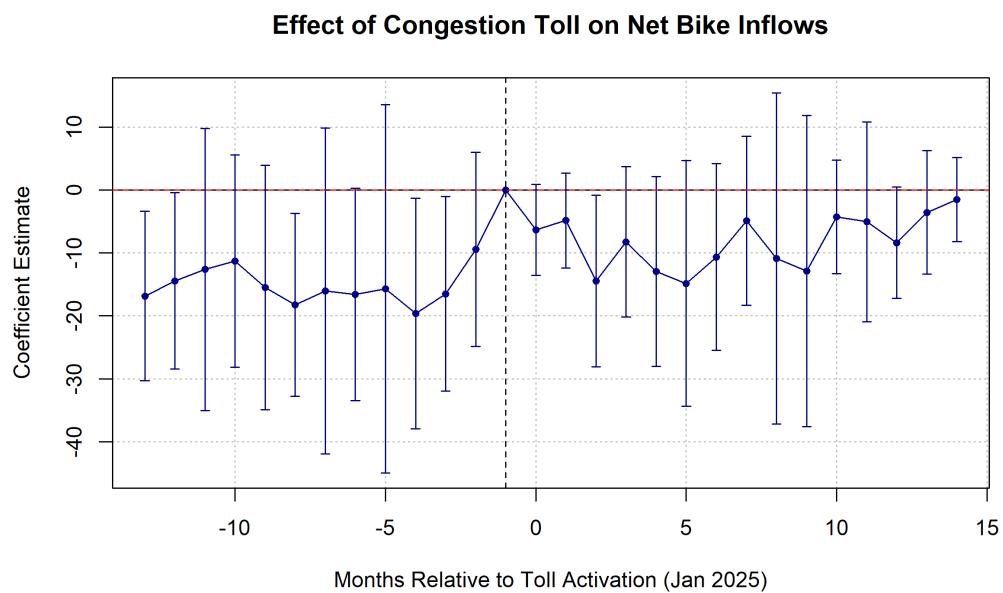


Figure 3: Enter Caption

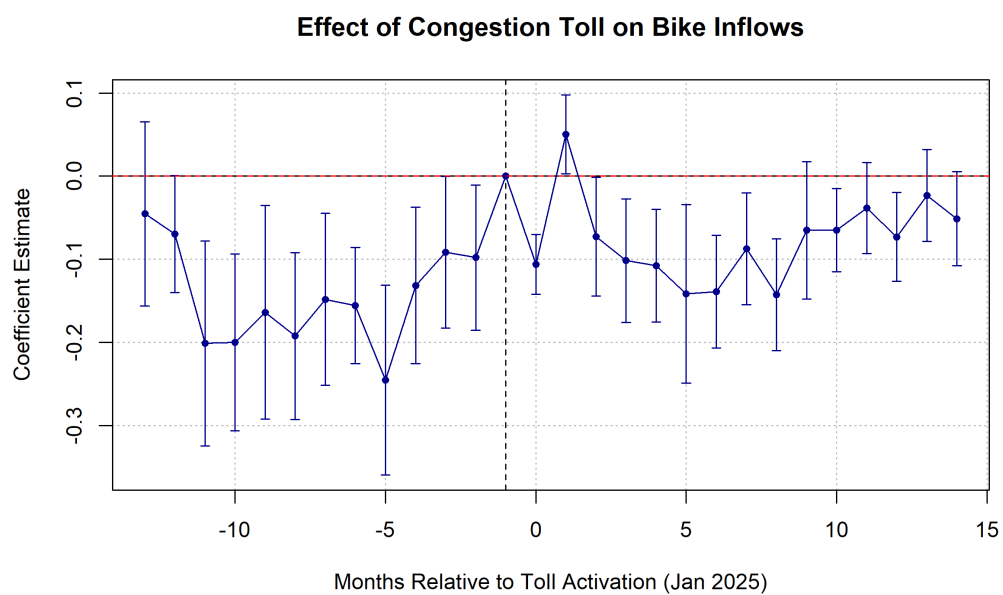


Figure 4: Enter Caption

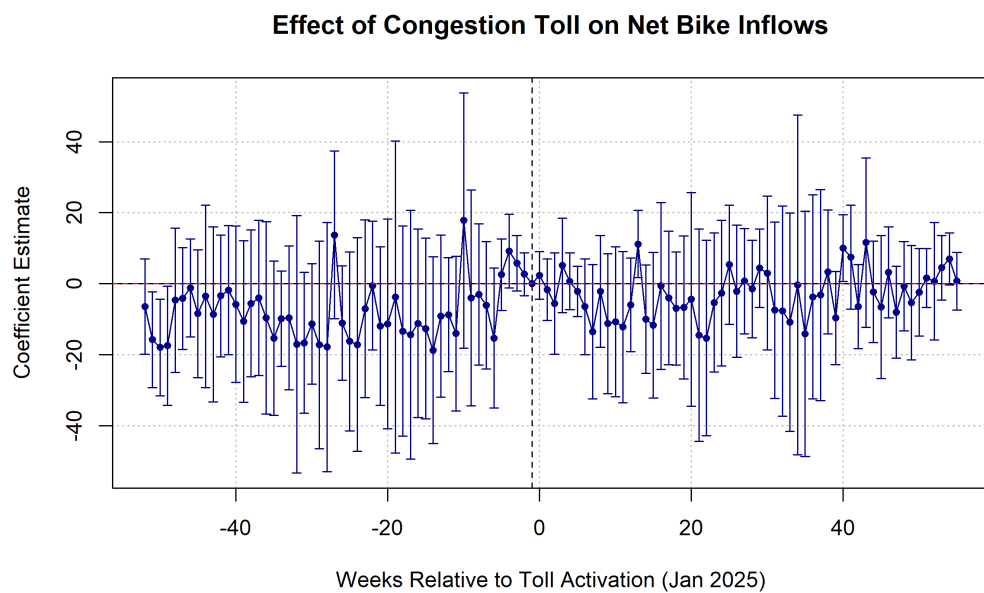


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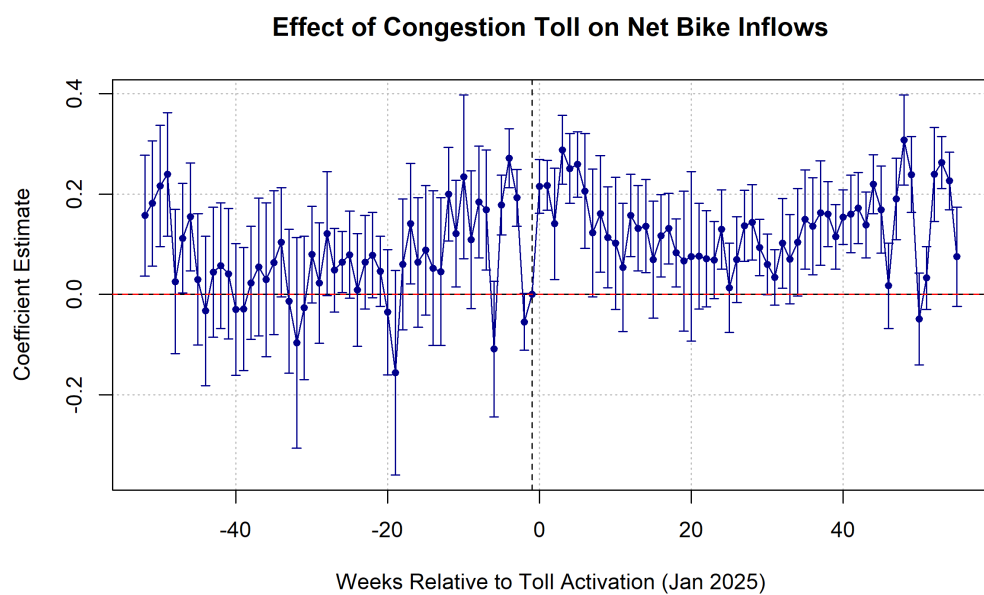


Figure 6: Enter Caption

Table 1: Daily Difference-in-Differences Estimates: Congestion Pricing Impact

| Dependent Variable:<br>Model: | Rides Started (Log)<br>(1) | Rides Ended (Log)<br>(2) | Net Inflows (Levels)<br>(3) |
|-------------------------------|----------------------------|--------------------------|-----------------------------|
| <i>Variables</i>              |                            |                          |                             |
| Toll Area $\times$ Post       | 0.0295<br>(0.0191)         | 0.0495***<br>(0.0154)    | 0.4397<br>(0.5052)          |
| <i>Fixed-effects</i>          |                            |                          |                             |
| station_id                    | Yes                        | Yes                      | Yes                         |
| date                          | Yes                        | Yes                      | Yes                         |
| <i>Fit statistics</i>         |                            |                          |                             |
| Observations                  | 465,276                    | 465,276                  | 465,276                     |
| R <sup>2</sup>                | 0.83102                    | 0.81556                  | 0.21864                     |
| Adjusted R <sup>2</sup>       | 0.83050                    | 0.81499                  | 0.21624                     |

*Clustered (station\_id) standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

Table 2: Daily Difference-in-Differences Estimates: Congestion Pricing Impact

| Dependent Variable:<br>Model: | Rides Started (Log)<br>(1) | Rides Ended (Log)<br>(2) | Rides Started (Levels)<br>(3) | Rides Ended (Levels)<br>(4) |
|-------------------------------|----------------------------|--------------------------|-------------------------------|-----------------------------|
| <i>Variables</i>              |                            |                          |                               |                             |
| Toll Area $\times$ Post       | 0.0295<br>(0.0191)         | 0.0495***<br>(0.0154)    | 0.4473<br>(0.9941)            | 0.8869<br>(1.047)           |
| <i>Fixed-effects</i>          |                            |                          |                               |                             |
| station_id                    | Yes                        | Yes                      | Yes                           | Yes                         |
| date                          | Yes                        | Yes                      | Yes                           | Yes                         |
| <i>Fit statistics</i>         |                            |                          |                               |                             |
| Observations                  | 465,276                    | 465,276                  | 465,276                       | 465,276                     |
| R <sup>2</sup>                | 0.83102                    | 0.81556                  | 0.78785                       | 0.78918                     |
| Adjusted R <sup>2</sup>       | 0.83050                    | 0.81499                  | 0.78720                       | 0.78853                     |

*Clustered (station\_id) standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

Table 3: Weekly Difference-in-Differences Estimates: Congestion Pricing Impact

| Dependent Variable:<br>Model: | Rides Started (Log)<br>(1) | Rides Ended (Log)<br>(2) | Rides Started (Levels)<br>(3) | Rides Ended (Levels)<br>(4) |
|-------------------------------|----------------------------|--------------------------|-------------------------------|-----------------------------|
| <i>Variables</i>              |                            |                          |                               |                             |
| Toll Area $\times$ Post       | 0.0168<br>(0.0257)         | 0.0620***<br>(0.0195)    | -0.8518<br>(7.057)            | 3.961<br>(7.322)            |
| <i>Fixed-effects</i>          |                            |                          |                               |                             |
| station_id                    | Yes                        | Yes                      | Yes                           | Yes                         |
| week_start                    | Yes                        | Yes                      | Yes                           | Yes                         |
| <i>Fit statistics</i>         |                            |                          |                               |                             |
| Observations                  | 59,514                     | 59,514                   | 59,514                        | 59,514                      |
| R <sup>2</sup>                | 0.90071                    | 0.79926                  | 0.87999                       | 0.85917                     |
| Adjusted R <sup>2</sup>       | 0.89941                    | 0.79663                  | 0.87841                       | 0.85733                     |

*Clustered (station\_id) standard-errors in parentheses*  
*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*